

Mouhannad Allouf

Senior Software Engineer • Systems Thinking • Blockchain • AI-Augmented Development

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 Damascus, Syria

Professional Summary

Senior Software Engineer who designs and ships reliable, scalable systems — not just code — and leverages AI-augmented workflows to deliver faster without cutting corners. Over a decade spent owning the full lifecycle of production systems across government platforms, fintech/payments and Web3: requirements, architecture, data migration & integration, deployment, and long-term operation. Holds an MSc in Big Data Systems (thesis on a **blockchain-based governance model for public services**) and approaches every project by first mapping how data, people and processes connect across the organization — then builds the system that fits.

Key Achievements

Systems Integration Blockchain / Web3 Data Migration & ETL Stakeholder Management Full-Stack & DevOps

- Consolidated **3 incompatible legacy fleet-tracking platforms into 1 unified system** for a national-scale operator (SaudiEX) — designing the full extraction-to-validated-schema pipeline so no live vehicle data is lost in the process.
- Architected a **dual-currency payment layer** (5 fiat gateways + crypto) across 4 countries for a regional e-commerce marketplace, unifying gateway-specific quirks behind one internal API so adding a new payment method is a config change, not a rewrite.
- Owned requirements-to-release for enterprise modules inside a national oil-sector information system for **3 years (2023–2026)** — the technical point of contact translating business need into shipped, tested functionality.
- Delivered **10+ full-stack production systems** for Syrian government ministries (agriculture, commerce, engineering placement); several remain in active daily use after 5+ years.
- Took blockchain from a Master's thesis into production: smart contracts live on **3 chains** (Ethereum, BNB Chain, Solana) plus enterprise Hyperledger deployments.
- Engineered hospital-grade **PACS/RIS/EMR interoperability** (DICOM/HL7), connecting real CT/MRI/US imaging modalities into a single diagnostic workflow.

Experience

Data Migration Engineer

20/6/2026 – Present

[SaudiEX](#) · Remote

Own the end-to-end data migration bridging three incompatible legacy fleet-tracking platforms into one unified system — systems-integration work that has to get schema, referential integrity and edge cases right the first time, since live vehicle tracking for the operator depends on it. Designed and executed a Python/Selenium ETL pipeline automating extraction, transformation, validation and loading across large-scale fleet datasets.

- Built automated extraction scripts using Python and Selenium to scrape structured data from legacy web UIs (vehicles, drivers, geofences, trailers, passengers, user accounts, alarm configurations, company hierarchies, and asset groups).
- Designed data mapping and transformation rules to reconcile schema differences across three source platforms, including field normalization, referential integrity checks, and conflict resolution for duplicate/overlapping records (IMEI, SIM, device assignments).

- Implemented a multi-step validation pipeline: row-count reconciliation, relationship integrity checks, and automated SQL schema generation for the target database.
- Produced comprehensive migration deliverables: CSV exports, JSON relationship maps, integrity audit reports, and a ready-to-deploy PostgreSQL schema — all versioned and documented for stakeholder review.
- Extracted administrative master data (company trees, account permissions, monitoring configurations) via authenticated session scraping with cookie persistence and pagination handling.
- Migrated data now powers live production systems recruiters can verify directly: [fleet tracking map](#) and [IVMS monitoring portal](#).
- Applied AI-augmented development across the migration: Claude (Code/Chat) and GitHub Copilot used to author and review extraction scripts, validation queries and migration documentation — faster delivery with human review at every step.

Python

Selenium

ETL

PostgreSQL

Data Validation

Web Scraping

Backend Developer

1/6/2024 – 1/3/2026

[Middle East Commerce](#)

· Remote

Backend engineer owning the payments layer for a multi-vendor e-commerce marketplace serving the GCC and Middle East — infrastructure where a single gateway outage blocks every seller's payout, so reliability and reconciliation came first. Designed and implemented payment interfaces and gateway integrations supporting both fiat and cryptocurrency transactions, unifying five-plus regional providers behind one internal API so onboarding a new payment method is a configuration change, not a new integration.

- Built payment integration layer supporting multiple fiat gateways (Stripe Connect, PayTabs, HyperPay, Moyasar, Tap Payments) and local payment methods (Mada, STC Pay, SADAD, KNET, COD) across Saudi Arabia, UAE, Kuwait, and Egypt.
- Integrated crypto payment processing via NOWPayments gateway, enabling BTC, ETH, and USDT transactions for platform fees and subscriptions.
- Implemented a 3-balance custodial wallet model (available / pending / locked) with Stripe Checkout fiat top-ups, refund handling via webhooks, and full audit trail with balance-before/balance-after tracking.
- Developed dual-flow payment architecture: Flow A (buyer-to-seller fiat escrow with commission deduction) and Flow B (user-to-platform fees accepting fiat or crypto with MEEC token discount).
- Built webhook handlers for payment provider IPN notifications, ensuring reliable transaction state synchronization and reconciliation.

Laravel

PHP

Stripe

NOWPayments

MySQL

REST API

Deputy Head — Technical Affairs (Investment & Development)

2023 – 1/6/2026

[Takamol Company](#)

· Damascus, Syria

Member of the company's business management team, responsible for specific modules of the core enterprise information system — from business need to live operation, in coordination with the system's owner company, Mahroukat (Ministry of Oil). The role sits at the intersection of business, engineering and operations — translating stakeholder needs into shipped functionality while keeping a critical oil-sector system stable in production.

- Requirements and specifications: analyze business needs and translate them into clear functional and technical specs. Author the requirement documents and prioritize the development backlog based on operational impact.
- Ticket management: open development tickets, follow them up with the owner company end-to-end, review delivered work against the spec, and keep stakeholders updated on progress.
- Quality assurance: design and execute test scenarios on the staging environment, approve releases to production, return tickets with detailed feedback when needed, and monitor stability after each release.
- User support: run the support channels (Telegram / WhatsApp) used by system end-users. Triage incoming reports, communicate with users, and convert validated issues into actionable tickets.
- Reporting and data extraction: write SQL queries directly against the production database to produce custom reports not exposed by the system UI, and propose new built-in reports for recurring needs.
- Data quality and reconciliation: detect and investigate duplicate or conflicting records via SQL, and reconcile data between system modules and the owner company's records.

- Cross-team coordination: act as the liaison between internal departments, the owner company (Mahroukat), and senior management. Unify requirements from different departments into a single integrated development plan.

Requirements Analysis

QA & Testing

SQL

MySQL

Stakeholder Management

Blockchain Developer

2021 – 2023

[HIVE Technologies](#)

· Remote

Built a multi-network Web3 wallet using JavaScript, Web3.js and MultiChain APIs, supporting account management and transaction signing across several blockchain networks from a single interface.

Contributed to a Hyperledger-based wheat supply-chain tracking platform. Wrote Go chaincode (smart contracts), set up the Docker / YAML deployment configuration, and helped model the on-chain data flow between farmers, mills and government inspectors.

Hyperledger

Web3.js

Go

Docker

Ethereum

Android Native & Web3 Developer

2021 – Nov 2025

[Incentives Company](#)

· Remote

PumpBNB — BNB Chain Token Launchpad: a no-code platform that lets anyone create and trade their own meme tokens on BNB Chain. Newly launched tokens trade against an automated bonding-curve pricing model until they reach a market-cap threshold, at which point liquidity is migrated to PancakeSwap and the token "graduates" to open trading. Designed and implemented the smart contracts, the trading API, and the web trading interface end-to-end.

FlipCoin — Provably-Fair Coin Flip on Solana: a peer-to-peer betting game where two players wager on a coin flip whose outcome is mathematically verifiable on-chain. Neither the players nor the platform can predict or influence the result. Built the on-chain Solana program using a commit-reveal protocol with verifiable randomness, plus the web client and the real-time game lobby.

AI Voice-Driven Game (Unity mobile + Wear OS companion): a Unity-based Android game with live voice interaction, where in-game NPCs listen to the player and respond in real time with dynamic, context-aware dialogue. Built the OpenAI integration layer (Realtime API for streaming voice, Whisper for transcription, Chat Completion for dialogue logic), the native Android ↔ Unity bridge plugin (Java), and a Wear OS smartwatch companion app that syncs game state and player stats with the phone over the Wearable Data API. Stack spans Unity / C#, Android / Java / Kotlin, and OpenAI APIs.

Solidity

Rust/Anchor

Next.js

React

Hardhat

OpenAI API

Unity

Kotlin

Android

Wear OS

Full-Stack Software Engineer

2015 – 2025

[HIAST \(Higher Institute of Applied Sciences & Technology\)](#)

· Damascus, Syria

- Livestock Management CMS — Ministry of Agriculture: full-stack management system tracking livestock records, vaccinations, and farm registries across governorates, with role-based workflows for field veterinarians and central administration. Designed the database schema, built the vaccination-scheduling logic and the statistical reports used by the central directorate. ASP.NET, C#, SQL Server.
- Census Data & Planning System — Ministry of Agriculture: data-collection and planning platform supporting the national agricultural census and resource planning across regions. Designed the data model and the entry forms, built the planning reports, and developed ETL routines to bring previous census rounds from legacy export files into the new Oracle schema — mapping, validation and reconciliation included. Java JSF, PrimeFaces, Oracle DB.
- Job Assignment System for Graduate Engineers: information system managing the placement of newly graduated engineers in public-sector positions, including eligibility, preferences, and ranking workflows. Implemented the eligibility rules, the candidate-ranking logic and the official assignment reports. Oracle DB, Java / PrimeFaces.
- Government Lab & Certificate-Issuing Systems — Ministry of Commerce: lab sample tracking, test results, and automated certificate-issuing workflows that significantly improved turnaround time for government services. Built the sample-registration and results-entry screens and the automated certificate generation and printing. Java JSF, PrimeFaces, Oracle.
- GPS Fleet Tracking System: end-to-end vehicle tracking platform — Android client apps for drivers, Java backend services, ExtJS web dashboard for live tracking and trip history, and a REST API tying it all together. Onboarding existing customers meant migrating their fleets from previous tracking platforms — mapped and transformed customer, vehicle, device and

driver records, imported them in controlled batches, and resolved duplicate / conflicting records (IMEI, SIM and device assignments) before go-live.

- Stock & Inventory Management System: warehouse and inventory management built on Oracle Database with Oracle Developer tooling — stock-movement workflows, item catalogues, and stocktaking and audit reports.
- Fleet Management Web Dashboard — Angular + Django: modern full-stack redesign of the fleet tracking web interface. Built the frontend SPA with Angular (TypeScript, RxJS, Angular Material) featuring real-time vehicle map views, trip history playback, driver management, and geofence configuration. Developed the Django REST backend with PostgreSQL, implementing JWT authentication, role-based access control, and RESTful APIs for vehicle telemetry, driver assignments, and reporting modules. Containerized the stack with Docker for staging and production deployment.

Angular

Django

TypeScript

Python

PostgreSQL

Docker

REST API

Data Migration / ETL

PL/SQL

Java

Oracle

SQL Server

Full-Stack Developer — Healthcare Systems (PACS / RIS / EMR)

2014 – 2016

Invest for All · Damascus, Syria

End-to-end radiology informatics and PACS/RIS integration. Researched, deployed and integrated **dcm4chee** (open-source PACS) for hospital imaging workflows. Engineered full interoperability between PACS, RIS and EMR via DICOM and HL7 standards. Implemented Modality Worklist (MWL), HL7 ORM order insertion, ORU results reporting, and bi-directional study status sync. Led technical evaluation of DICOM storage SCP/SCU, WADO, and IHE profiles.

Configured production-ready PACS on Linux with JBoss/WildFly, connected real modalities (CT, MRI, US), validated image archiving and study routing. Developed custom web admin modules, DICOM viewers, and reporting dashboards around the healthcare stack. Produced complete deployment runbooks and integration prototypes for hospital-grade rollout.

dcm4chee

PACS

RIS

EMR

DICOM

HL7

Modality Worklist

JBoss/WildFly

Java

Linux

Technical Skills

Blockchain

Solana, Ethereum, BNB Chain, Hyperledger, Anchor, Hardhat, Web3.js

Frontend

Angular, React, Next.js, TypeScript, Tailwind, Wagmi, Zustand

Data & Migration

ETL, data mapping & transformation, validation & reconciliation, PL/SQL, batch imports, rollback & recovery

Mobile

Android, Kotlin, Unity

Languages

TypeScript, JavaScript, Java, Kotlin, C#, Go, Python, PHP, Laravel, Rust, Solidity, SQL

Backend & DB

Django, Node.js, Express, ASP.NET, Laravel, PHP, PostgreSQL, MySQL, Oracle, Redis, Prisma

DevOps & Tools

Docker, Git, GitLab CI, Linux, Vercel, Render

AI-Augmented Development

Claude (Code/Chat) & GitHub Copilot for delivery, AI-assisted architecture & code review, prompt engineering for dev workflows, AI-assisted test generation & documentation

Education

MSc Big Data Systems

[HIAST](#) · 2018 – 2021

Thesis: **blockchain-based governance model for public services**

BSc Computer Science & IT

[HIAST](#) · 2005 – 2010

Certifications

Blockchain Specialization	Univ. at Buffalo / Coursera	2021 · Verify
Kotlin for Java Developers	JetBrains / Coursera	2021 · Verify

Languages

- Arabic — Native
- English — Professional working proficiency (daily working language)
- French — Reading

Personal

Born: May 22, 1985

Nationality: Syrian

Status: Married

Let's build something together.

I help organizations design reliable, scalable systems that leverage AI to solve real business problems.